



US012414670B2

(12) **United States Patent**
Ryan et al.

(10) **Patent No.:** **US 12,414,670 B2**
(45) **Date of Patent:** **Sep. 16, 2025**

(54) **SYSTEMS FOR CADDY POSITIONING IN DISHWASHER APPLIANCES**

(71) Applicant: **Haier US Appliance Solutions, Inc.**,
Wilmington, DE (US)

(72) Inventors: **Justin Daniel Ryan**, Louisville, KY
(US); **Steven Chadwick Koepke**,
LaGrange, KY (US); **Nathaniel Lee**
Min Sheeran, Louisville, KY (US)

(73) Assignee: **Haier US Appliance Solutions, Inc.**,
Wilmington, DE (US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 154 days.

(21) Appl. No.: **18/314,563**

(22) Filed: **May 9, 2023**

(65) **Prior Publication Data**

US 2024/0374114 A1 Nov. 14, 2024

(51) **Int. Cl.**
A47L 15/50 (2006.01)
A47L 15/16 (2006.01)
A47L 15/22 (2006.01)

(52) **U.S. Cl.**
CPC **A47L 15/505** (2013.01); **A47L 15/16**
(2013.01); **A47L 15/22** (2013.01)

(58) **Field of Classification Search**
CPC A47L 15/16; A47L 15/22; A47L 15/4278;
A47L 15/50; A47L 15/502; A47L 15/505;
A47L 15/508
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

10,213,086 B2 2/2019 Blanchard et al.
11,375,874 B2 7/2022 Feddema et al.

2005/0045215 A1* 3/2005 Raches A47L 15/502
134/179
2012/0097198 A1 4/2012 Schmidt et al.
2018/0110396 A1* 4/2018 Fawaz A47L 15/4214
2021/0330172 A1 10/2021 Pugh et al.
2022/0133124 A1 5/2022 Kim et al.
2022/0142452 A1 5/2022 Dietrich et al.
2022/0287543 A1 9/2022 Feddema et al.

FOREIGN PATENT DOCUMENTS

AT 12122 U1 11/2011
CN 215502876 U 1/2022
CN 114305277 A 4/2022
EP 3020317 B1 3/2018
WO WO2016155797 A1 10/2016

OTHER PUBLICATIONS

FREEFLEXTM Third Rack Dishwasher. KitchenAid. (n.d.). <https://www.kitchenaid.com/major-appliances/dishwashers/third-rack.html>, accessed May 8, 2023, 17 pages.

* cited by examiner

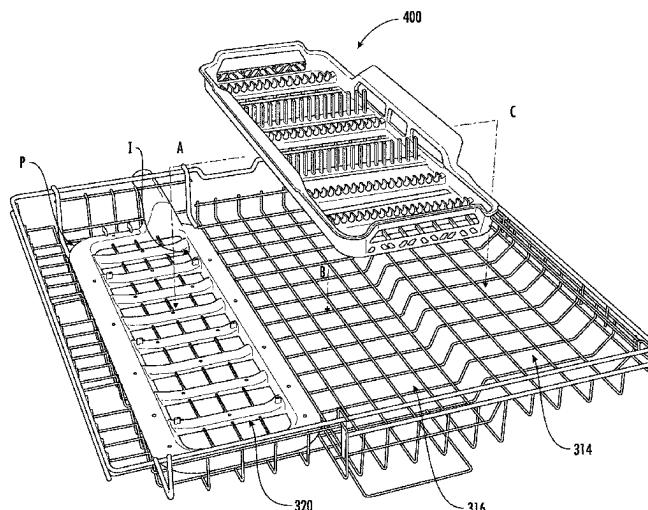
Primary Examiner — David G Cormier

(74) *Attorney, Agent, or Firm* — Dority & Manning, P.A.

(57) **ABSTRACT**

A dishwasher appliance includes a tub that defines a wash chamber, a third rack assembly slidably positioned in the wash chamber above a second rack assembly, and a third spray assembly positioned within the third rack assembly. The third spray assembly is configured to direct wash fluid at articles located in the third rack assembly. The dishwasher appliance also includes a caddy that is positionable on the third spray assembly. The caddy includes a plurality of mounting projections at a bottom side of the caddy. The mounting projections are configured to engage the third spray assembly in order to locate the caddy on the third spray assembly.

16 Claims, 7 Drawing Sheets



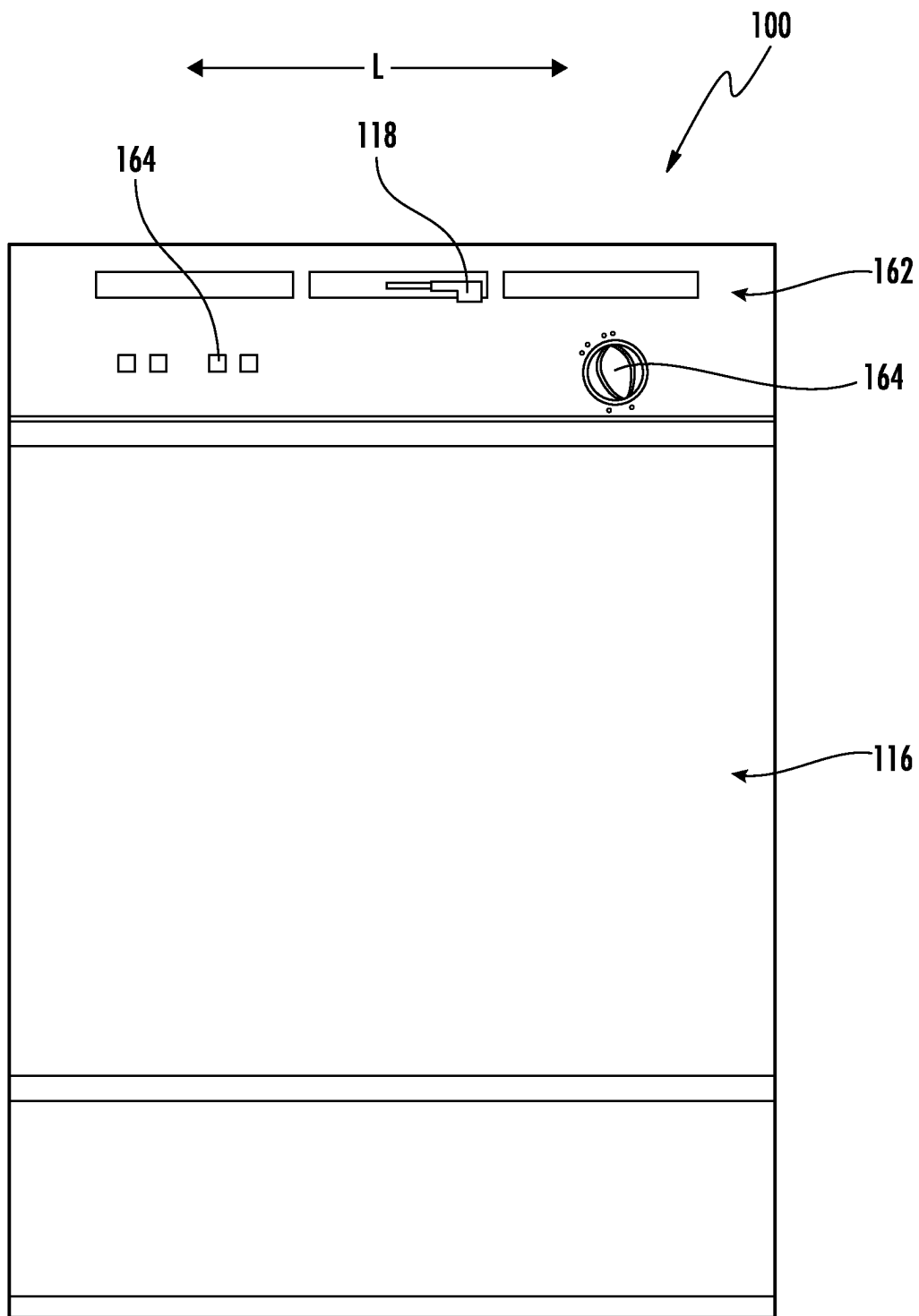


FIG. 1

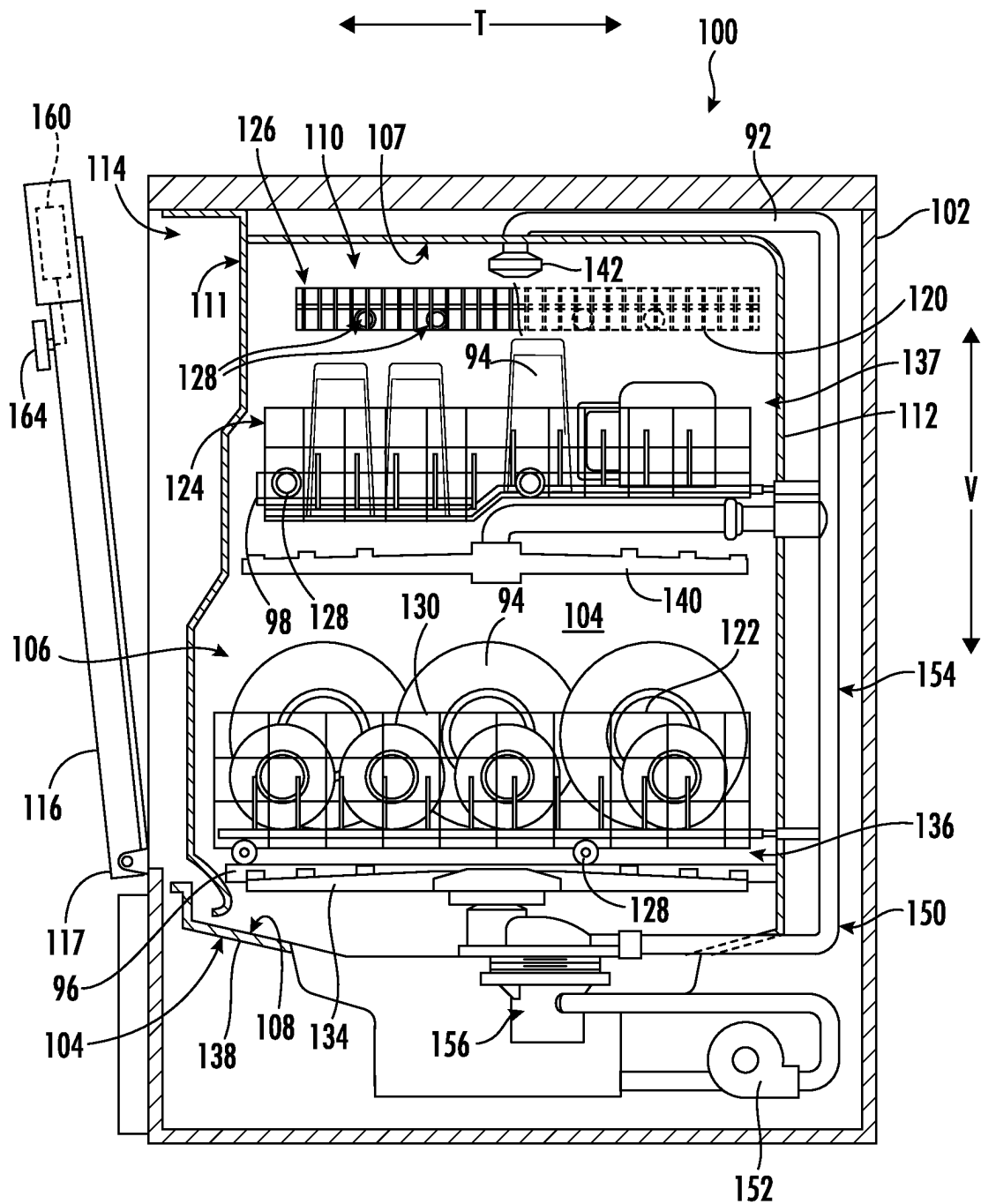


FIG. 2

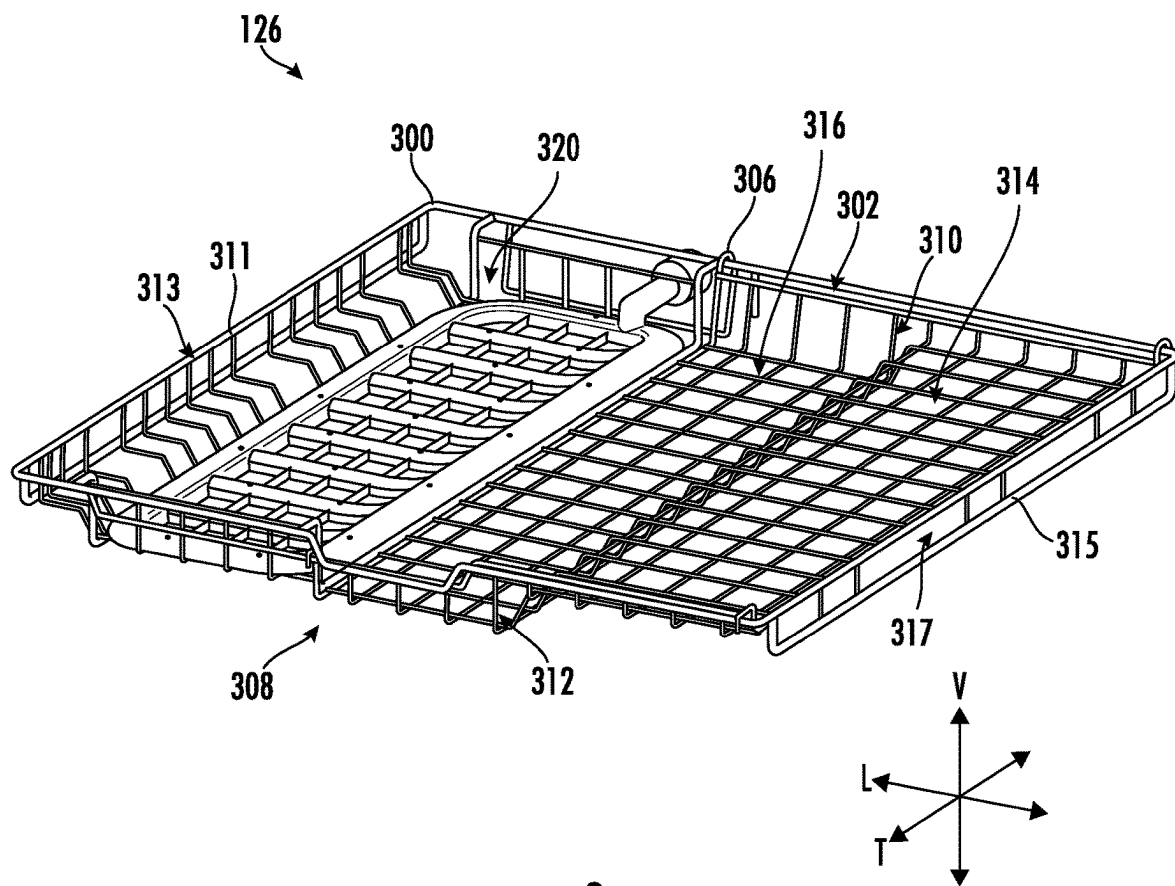
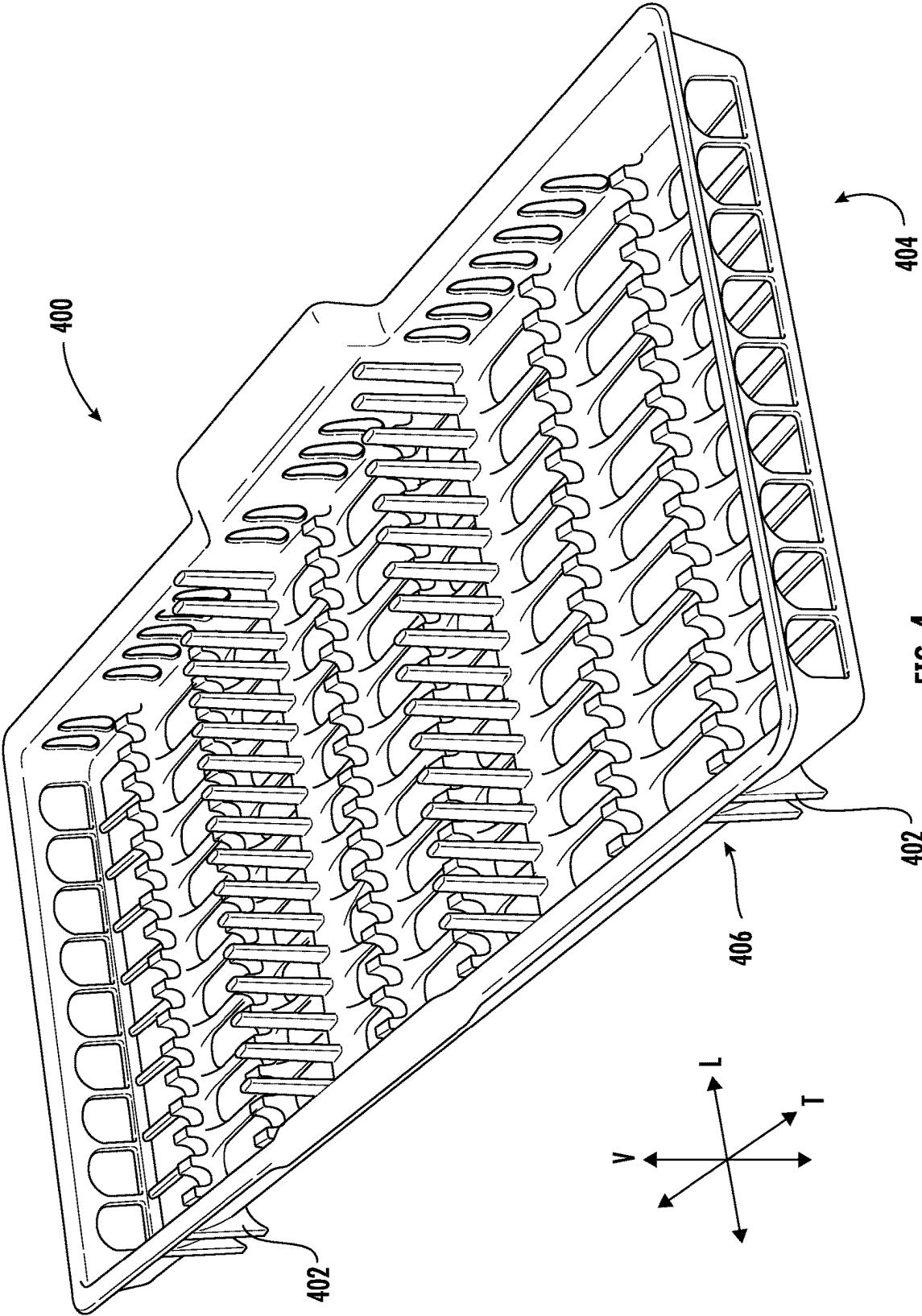


FIG. 3



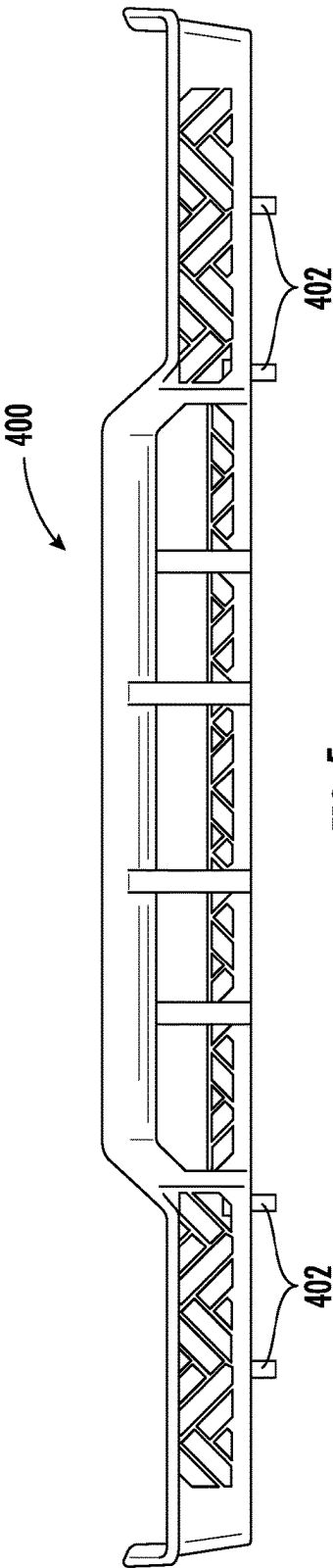


FIG. 5

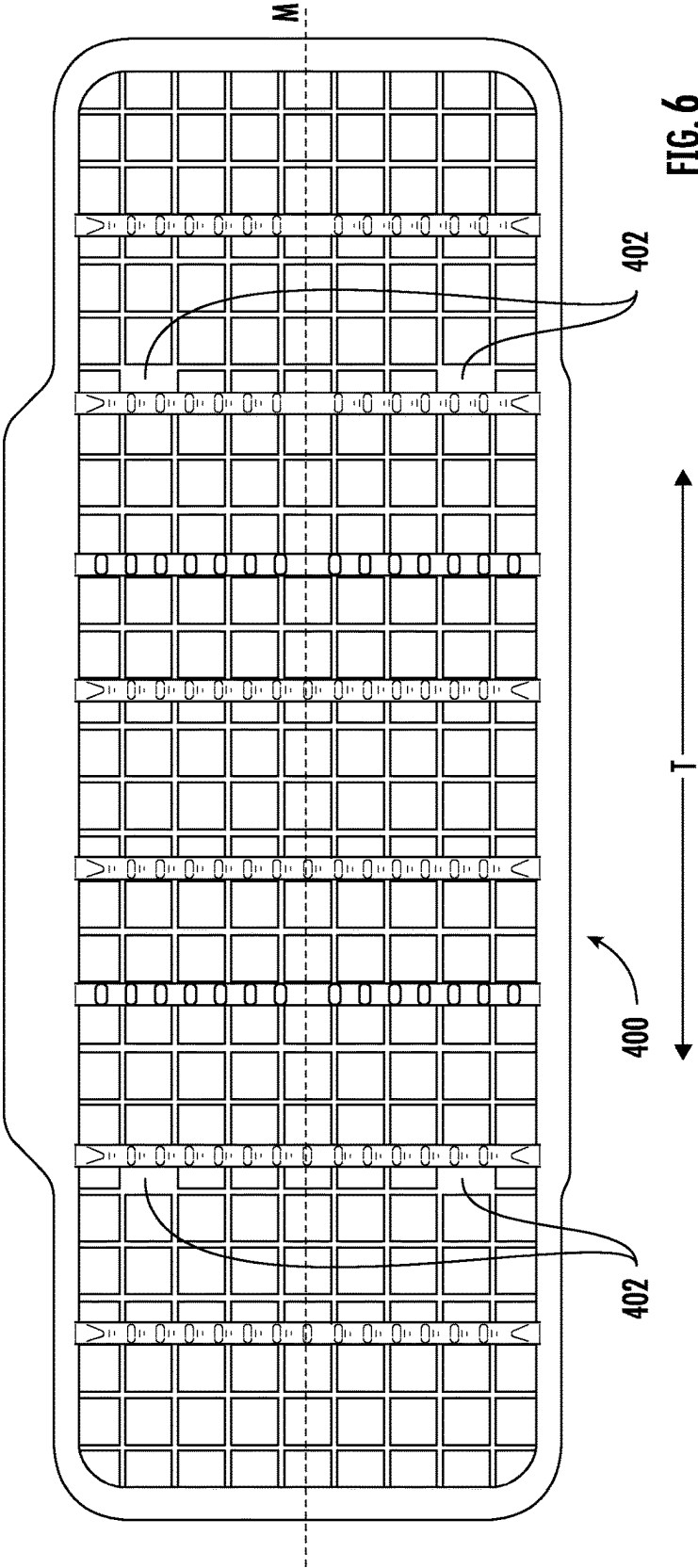
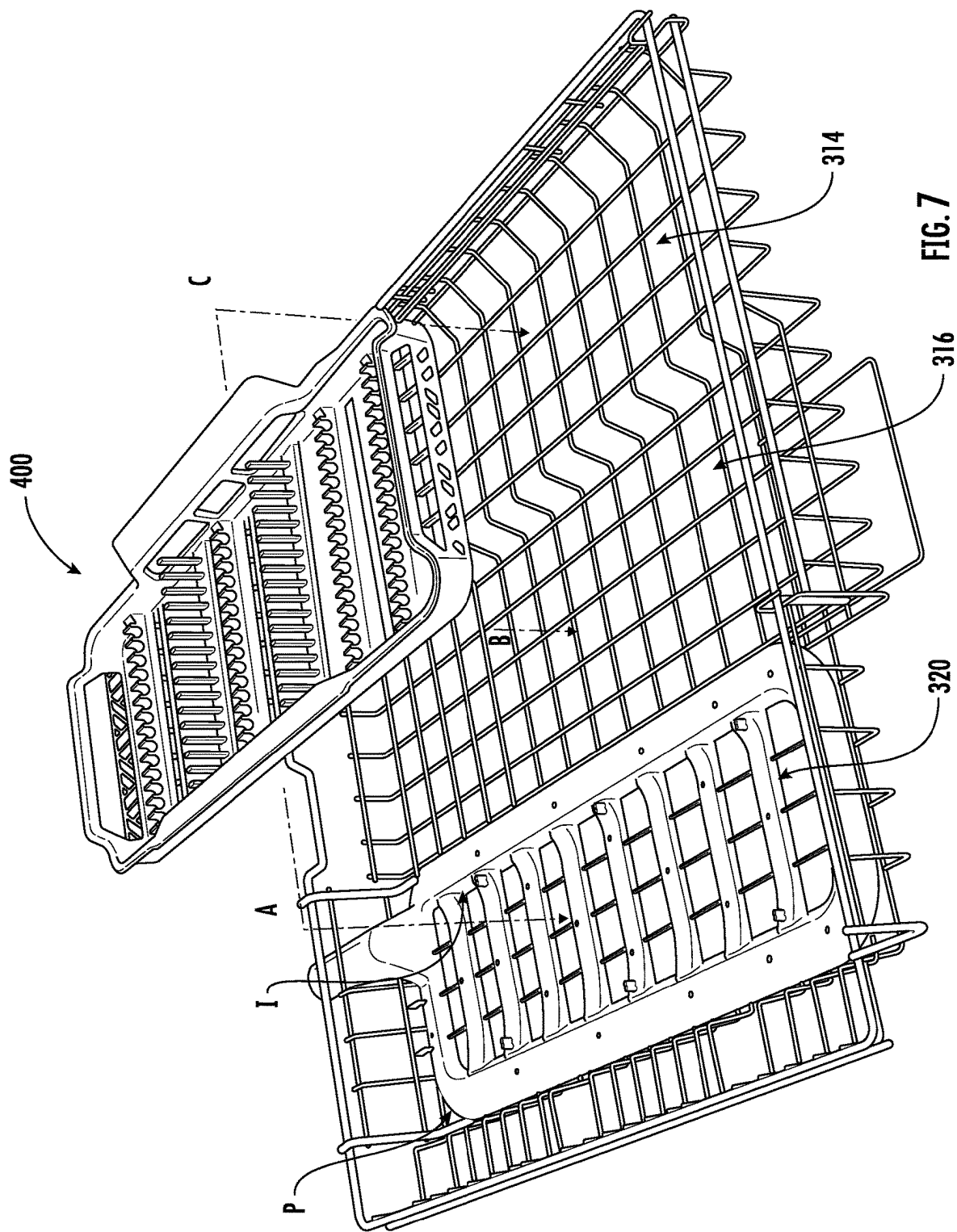
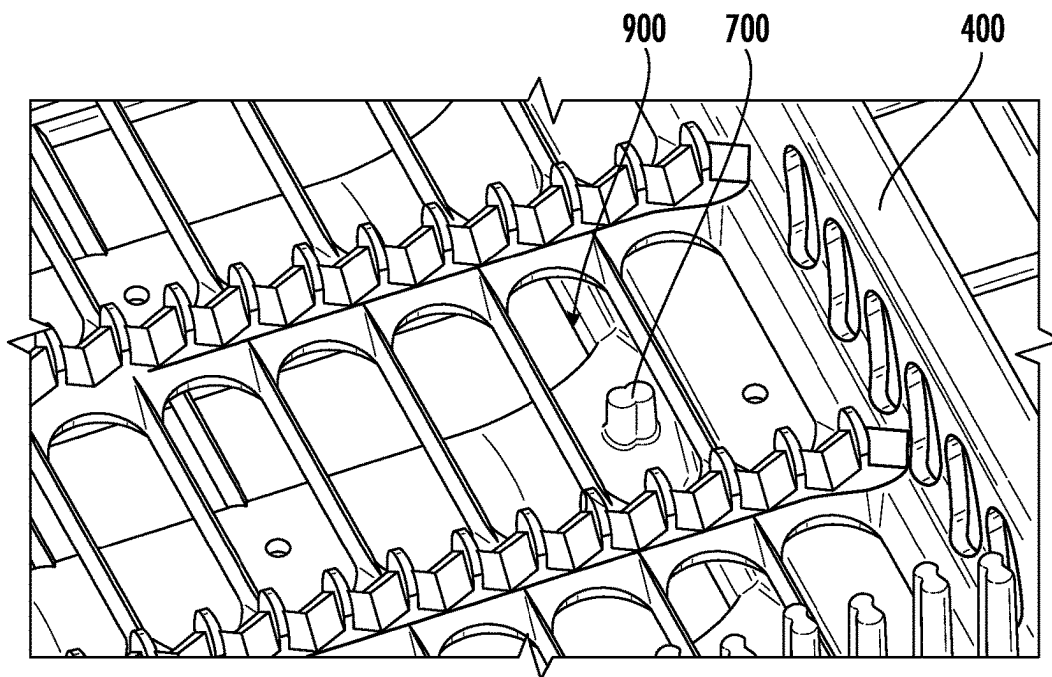
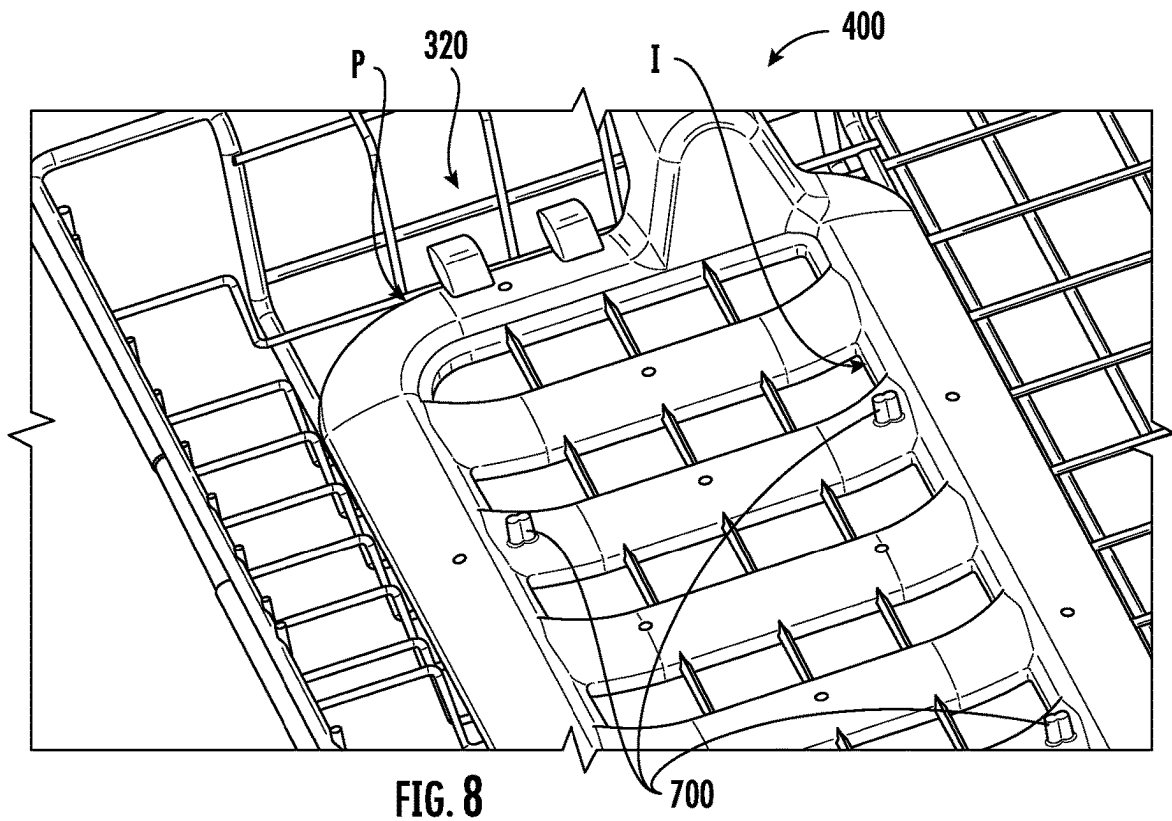


FIG. 6





1

SYSTEMS FOR CADDY POSITIONING IN DISHWASHER APPLIANCES

FIELD OF THE INVENTION

The present subject matter relates generally to caddy positioning in dishwasher appliances, particularly locating a caddy in a third rack.

BACKGROUND OF THE INVENTION

Dishwasher appliances generally include rack assemblies for positioning various articles for cleaning within a wash chamber. One or more devices, such as nozzles or spray assemblies, may be included at various locations relative to the rack assemblies for purposes of delivering fluids as part of the cleaning process. During the cleaning cycle, the rack assemblies can support and position the articles while also having openings that allow fluid to pass through to the articles. Factors, such as the velocity of the fluid, orientation of the fluid spray or stream relative to the articles, the shape and density of the articles in the rack assemblies, and others, can impact the effectiveness of the cleaning cycle.

One or more rack assemblies may be used in dishwasher appliances for user convenience. Multiple rack assemblies on multiple levels within dishwasher appliances may allow users to place articles of differing heights and sizes in optimal positions to allow for proper cleaning of the articles. Some dishwasher appliances may include one or more baskets which may also be provided for holding articles, particularly smaller or for more narrow articles, such as silverware. Additionally, the user may have the option of, e.g., placing articles, such as silverware, within a basket on a lower rack assembly or placing the silverware directly (without the basket) onto an upper rack assembly specially configured for the receipt of such articles.

The positioning of articles within a dishwasher appliance can affect the fluid dynamics to which the articles are exposed during the cleaning process. For example, articles placed in a lower rack assembly may be subjected to different spray assemblies with different spray patterns, velocities, and spray duration than articles placed in a higher rack assembly. As dishwasher appliances have begun including more racks, space and positioning of dishwashing components inside of the dishwasher is important to ensure proper washing.

BRIEF DESCRIPTION OF THE INVENTION

Aspects and advantages of the invention will be set forth in part in the following description, or may be apparent from the description, or may be learned through practice of the invention.

In one example embodiment, a dishwasher appliance defines a vertical direction, a lateral direction, and a transverse direction. The vertical, lateral, and transverse directions are mutually perpendicular. The dishwasher appliance includes a tub that defines a wash chamber, and a first rack assembly slidably positioned within the wash chamber. The dishwasher appliance also includes a first spray assembly positioned in the wash chamber and configured to direct wash fluids at the first rack assembly, as well as a second rack assembly slidably positioned in the wash chamber above the first rack assembly. The second spray assembly is positioned in the wash chamber and configured to direct wash fluids at the second rack assembly. The dishwashing appliance further includes a third rack assembly slidably

2

positioned in the wash chamber above the second rack assembly, and a third spray assembly positioned within the third rack assembly. The third spray assembly is configured to direct wash fluid at articles located in the third rack assembly. The dishwasher appliance also includes a caddy that is positionable on the third spray assembly. The caddy includes a plurality of mounting projections at a bottom side of the caddy. The mounting projections are configured to engage the third spray assembly in order to locate the caddy on the third spray assembly.

In another example embodiment, a dishwasher appliance defines a vertical direction, a lateral direction, and a transverse direction. The vertical, lateral, and transverse directions are mutually perpendicular. The dishwasher appliance includes a tub that defines a wash chamber, and a first rack assembly slidably positioned within the wash chamber. The dishwasher appliance also includes a first spray assembly positioned in the wash chamber and configured to direct wash fluids at the first rack assembly, as well as a second rack assembly slidably positioned in the wash chamber above the first rack assembly. The second spray assembly is positioned in the wash chamber and configured to direct wash fluids at the second rack assembly. The dishwashing appliance further includes a third rack assembly slidably positioned in the wash chamber above the second rack assembly, and a third spray assembly positioned over the third rack assembly. The third spray assembly is configured to direct wash fluid at articles located in the third rack assembly. The third spray assembly includes a plurality of posts extending in the vertical direction. The dishwasher appliance also includes a caddy positionable on the third spray assembly. The caddy includes a plurality of mounting projections at a bottom side of the caddy. The mounting projections are configured to engage the third spray assembly in order to locate the caddy on the third spray assembly. The plurality of posts of the third spray assembly are configured to restrict movement of the caddy in the lateral and transverse direction.

These and other features, aspects and advantages of the present invention will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

BRIEF DESCRIPTION OF THE DRAWINGS

A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures.

FIG. 1 provides a perspective view of an example embodiment of a dishwashing appliance of the present disclosure.

FIG. 2 provides a side, cross sectional view of the example dishwashing appliance of FIG. 1.

FIG. 3 provides a perspective view of an example embodiment of a rack of the dishwashing appliance of FIG. 1.

FIG. 4 provides a perspective view of an example embodiment of a caddy in accordance with aspects of the present disclosure.

FIG. 5 provides a side view of an example embodiment of a caddy in accordance with aspects of the present disclosure.

FIG. 6 provides a bottom view of the example caddy of FIG. 5.

3

FIG. 7 provides a schematic view of the positioning of a caddy in the rack of FIG. 3 according to aspects of the present disclosure.

FIG. 8 provides a perspective view of an example spray assembly according to aspects of the present disclosure.

FIG. 9 provides a perspective view of the spray assembly of FIG. 8 engaging with an example caddy according to aspects of the present disclosure.

Repeat use of reference characters in the present specification and drawings is intended to represent the same or analogous features or elements of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope or spirit of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

As used herein, the terms “includes” and “including” are intended to be inclusive in a manner similar to the term “comprising.” Similarly, the term “or” is generally intended to be inclusive (i.e., “A or B” is intended to mean “A or B or both”). Approximating language, as used herein throughout the specification and claims, is applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about,” “approximately,” and “substantially,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value. For example, the approximating language may refer to being within a 10 percent margin.

As used herein, the term “article” may refer to, but need not be limited to dishes, pots, pans, silverware, and other cooking utensils and items that can be cleaned in a dishwashing appliance. The term “wash cycle” is intended to refer to one or more periods of time during which a dishwashing appliance operates while containing the articles to be washed and uses a detergent and water, preferably with agitation, to e.g., remove soil particles including food and other undesirable elements from the articles. The term “rinse cycle” is intended to refer to one or more periods of time during which the dishwashing appliance operates to remove residual soil, detergents, and other undesirable elements that were retained by the articles after completion of the wash cycle. The term “drain cycle” is intended to refer to one or more periods of time during which the dishwashing appliance operates to discharge soiled water from the dishwashing appliance. The term “cleaning cycle” is intended to refer to one or more periods of time that may include a wash cycle, rinse cycle, and/or a drain cycle. The term “wash fluid” refers to a liquid used for washing and/or rinsing the articles and is typically made up of water that may include other additives such as detergent or other treatments.

FIGS. 1 and 2 depict an example domestic dishwasher or dishwashing appliance 100 that may be configured in accor-

4

dance with aspects of the present disclosure. For the particular embodiment of FIGS. 1 and 2, the dishwasher appliance 100 includes a cabinet 102 (FIG. 2) having a tub 104 therein that defines a wash chamber 106 for receipt of articles 94 for washing. As shown in FIG. 2, tub 104 extends between a top 107 and a bottom 108 along a vertical direction V, between a pair of opposing side walls 110 along a lateral direction L, and between a front side 111 and a rear side 112 along a transverse direction T. Each of the vertical direction V, lateral direction L, and transverse direction T are mutually perpendicular to one another.

In this regard, as used herein, the terms “cabinet,” “housing,” and the like are generally intended to refer to an outer frame or support structure for appliance 100, e.g., including any suitable number, type, and configuration of support structures formed from any suitable materials, such as a system of elongated support members, a plurality of interconnected panels, or some combination thereof. It should be appreciated that cabinet 102 does not necessarily require an enclosure and may simply include open structure supporting various elements of appliance 100. By contrast, cabinet 102 may enclose some or all portions of an interior of cabinet 102. It should be appreciated that cabinet 102 may have any suitable size, shape, and configuration while remaining within the scope of the present subject matter.

The tub 104 includes a front opening 114 and a door 116 hinged at its bottom 117 for movement between a normally closed vertical position (shown in FIG. 1), wherein the wash chamber 106 is sealed shut for washing operation, and a horizontal open position for loading and unloading of articles from the dishwasher appliance 100. According to example embodiments, dishwasher appliance 100 further includes a door closure mechanism or assembly 118 that is used to lock and unlock door 116 for accessing and sealing wash chamber 106.

At least one rack assembly is slidably positioned within wash chamber 106 and is configured for the receipt of articles for cleaning. For the example embodiment shown in FIG. 2, opposing tub sidewalls 110 accommodate a plurality of rack assemblies. More specifically, guide rails 96, 98 and 120 may be mounted to (or formed as part of) sidewalls 110 for supporting a first rack assembly 122 (also referred to as a lower rack assembly 122), a middle rack assembly 124 (also referred to as a second rack assembly 124), and a third rack assembly 126 (also referred to as an upper rack assembly 126). As illustrated, upper rack assembly 126 is positioned at a top portion of wash chamber 106 above middle rack assembly 124, which is positioned above lower rack assembly 122 along the vertical direction V. Additional details regarding the upper rack assembly 126 will be provided herein. For this example embodiment, upper rack assembly 126 is supported on opposing sidewalls 110 by rails 120 but rails 120 could be supported on and attached to middle rack assembly 124 as well.

Each rack assembly 122, 124, 126 is adapted for movement along transverse direction T between an extended loading position (not shown) in which the rack is substantially positioned outside the wash chamber 106, and a retracted position (shown in FIGS. 1 and 2) in which the rack is located inside the wash chamber 106. This may be facilitated, for example, by rollers, or guide wheels 128 mounted onto rack assemblies 122, 124, 126, respectively. Although guide rails 96, 98, 120 and guide wheels 128 are illustrated herein as facilitating movement of the respective rack assemblies 122, 124, 126, it should be appreciated that any suitable sliding mechanism or member may be used according to alternative embodiments. In some embodi-

ments, dishwasher appliance **100** may accommodate a different number of rack assemblies and supporting guide rails. For example, dishwasher appliance **100** may accommodate only first rack assembly **122** and upper rack assembly **126**, with accompanying guide rails.

Some or all of the rack assemblies **122**, **124**, **126** may be fabricated into lattice, or grid pattern, structures including a plurality of wires or elongated members **130** (for clarity of illustration, not all elongated members making up rack assemblies **122**, **124**, **126** are shown in FIG. 2). The plurality of wires or elongated members **130** may be either steel and stainless steel, and the wire may be coated with none, one, or more of nylon and polyvinyl chloride. Rack assemblies **122**, **124**, **126** are generally configured for supporting articles **94** within wash chamber **106** while allowing a flow of wash fluid to reach and impinge on those articles, e.g., during a cleaning or rinsing cycle. For some embodiments, a silverware basket (not shown) is removably attached to a rack assembly, e.g., lower rack assembly **122**, for placement of silverware, utensils, and the like, that are otherwise too small or delicate to be accommodated by rack **122**.

At least one spray assembly is located in wash chamber **106** and is configured to direct wash fluids onto at least on rack assembly for washing articles located therein. For the example embodiment of FIG. 2, dishwasher appliance **100** further includes a plurality of spray assemblies for urging a flow of water or wash fluid onto the articles placed within wash chamber **106**. More specifically, as illustrated in FIG. 2, dishwasher appliance **100** includes a lower spray assembly **134** (also referred to as a lower spray arm assembly **134**) disposed in a lower region **136** of wash chamber **106** and above a sump **138** so as to rotate in relatively close proximity to lower rack assembly **122**. Similarly, a mid-level spray assembly **140** is located in an upper region **137** of wash chamber **106** and may be located below and in close proximity to middle rack assembly **124**. In this regard, mid-level spray arm assembly **140** may generally be configured for urging a flow of wash fluid up through middle rack assembly **124** and third rack assembly **126**. Additionally, an upper spray assembly **142** may be located above upper or third rack assembly **126** along the vertical direction V. In this manner, upper spray assembly **142** may be configured for urging and/or cascading a flow of wash fluid downward over rack assemblies **122**, **124**, and **126**.

The various spray assemblies and manifolds described herein may be part of a fluid distribution system or fluid circulation assembly **150** for circulating water and wash fluid in the tub **104**. More specifically, fluid circulation assembly **150** includes a pump **152** for circulating water and wash fluid (e.g., detergent, water, and/or rinse aid) in the tub **104**. Pump **152** may be located within sump **138** or within a machinery compartment located below sump **138** of tub **104**, as generally recognized in the art. Fluid circulation assembly **150** may include one or more fluid conduits or circulation piping for directing water and/or wash fluid from pump **152** to the various spray assemblies and manifolds. For example, as illustrated in FIG. 2, a primary supply conduit **154** may extend from pump **152**, along rear side **112** of tub **104** along the vertical direction V to supply wash fluid throughout wash chamber **106**. In some examples, a secondary supply conduit (not shown) may supply additional wash fluid to one or more various spray assemblies and manifolds.

As illustrated, primary supply conduit **154** is used to supply wash fluid to mid-level spray arm assembly **140** while a secondary supply conduit **92** supplies wash fluid to upper spray assembly **142**. Diverter assembly **156** can allow

selection between spray assemblies **134** and **140**, **142** being supplied with wash fluid. However, it should be appreciated that according to alternative embodiments, any other suitable plumbing configuration may be used to supply wash fluid throughout the various spray manifolds and assemblies described herein.

Each spray assembly **134**, **140**, **142** or other spray device may include an arrangement of discharge ports or orifices for directing wash fluid received from pump **152** onto dishes or other articles located in wash chamber **106**. The arrangement of the discharge ports, also referred to as jets, apertures, or orifices, may provide a rotational force by virtue of wash fluid flowing through the discharge ports. Alternatively, spray assemblies **134**, **140**, **142** may be motor-driven, or may operate using any other suitable drive mechanism. Spray manifolds and assemblies may also be stationary. Movement of the spray arm assemblies **134** and **140** and the spray from fixed manifolds like spray assembly **142** provides coverage of dishes, silverware, and other dishwasher contents and articles **94** to be cleaned with a washing spray. Other configurations of spray assemblies may be used as well. For example, dishwasher appliance **100** may have additional spray assemblies for cleaning silverware, for scouring casserole dishes, for spraying pots and pans, for cleaning bottles, etc. One skilled in the art will appreciate that the embodiments discussed herein are used for the purpose of explanation only and are not limitations of the present subject matter.

In operation, pump **152** draws wash fluid in from sump **138** and pumps it to a diverter assembly **156**, e.g., which is positioned within sump **138** of dishwasher appliance. Diverter assembly **156** may include a diverter disk (not shown) disposed within a diverter chamber (not shown) for selectively distributing the wash fluid to the spray assemblies **134**, **140**, **142** and/or other spray manifolds or devices. For example, the diverter disk may have a plurality of apertures that are configured to align with one or more outlet ports (not shown) at the top of diverter chamber (not shown). In this manner, the diverter disk may be selectively rotated to provide wash fluid to the desired spray device.

According to an example embodiment, diverter assembly **156** is configured for selectively distributing the flow of wash fluid from pump **152** to various fluid supply conduits, only some of which (e.g., **154**) are illustrated in FIG. 2 for clarity. More specifically, diverter assembly **156** may include four outlet ports (not shown) for supplying wash fluid to a first conduit for rotating lower spray arm assembly **134** in the clockwise direction, a second conduit for rotating lower spray arm assembly **134** in the counterclockwise direction, a third conduit for spraying rack assembly **126** (shown in FIG. 2) as a silverware rack, and a fourth conduit for supplying only mid-level and/or upper spray assemblies **140**, **142**. Other configurations of diverter assembly **156** and/or other components (e.g., valves) may be used to allow various choices in the operation of the spray assemblies **134**, **140**, and **142** during a cleaning cycle.

The dishwasher appliance **100** is further equipped with a controller **160** (FIG. 2) to regulate operation of the dishwasher appliance **100**. Controller **160** may include one or more memory devices and one or more microprocessors, such as general or special purpose microprocessors operable to execute programming instructions or micro-control code associated with a cleaning cycle. The memory may represent random access memory such as DRAM or read only memory such as ROM or FLASH. In one embodiment, the processor executes programming instructions stored in memory. The memory may be a separate component from

7

the processor or may be included onboard within the processor. Alternatively, controller 160 may be constructed without using a microprocessor, e.g., using a combination of discrete analog and/or digital logic circuitry (such as switches, amplifiers, integrators, comparators, flip-flops, AND gates, and the like) to perform control functionality instead of relying upon software.

The controller 160 may be positioned in a variety of locations throughout dishwasher appliance 100. In the illustrated embodiment, the controller 160 may be located within a control panel area 162 of door 116. In such an embodiment, input/output (“I/O”) signals may be routed between the control system and various operational components of dishwasher appliance 100 along wiring harnesses that may be routed through the bottom of door 116. Typically, the controller 160 includes a user interface panel/controls 164 (FIG. 1) through which a user may select various operational features and modes and monitor progress of the dishwasher appliance 100. In one embodiment, the user interface 164 may represent a general purpose I/O (“GPIO”) device or functional block. In one embodiment, the user interface 164 may include input components, such as one or more of a variety of electrical, mechanical, or electro-mechanical input devices including rotary dials, push buttons, and touch pads. The user interface 164 may include a display component, such as a digital or analog display device designed to provide operational feedback to a user. The user interface 164 may be in communication with the controller 160 via one or more signal lines or shared communication busses.

It should be appreciated that the invention is not limited to any particular style, model, or configuration of dishwasher appliance 100. The example embodiment depicted in FIGS. 1 and 2 is for illustrative purposes only. For example, different locations may be provided for user interface 164, different configurations, including providing one or more rack assemblies 122, 124, 126 and one or more spray assemblies 134, 140, 142, to dishwasher appliance 100 may be used, different configurations may be provided for rack assemblies 122, 124, 126, different spray assemblies 134, 140, 142 and spray manifold configurations may be used, and other differences may be applied while remaining within the scope of the present subject matter.

Shown in FIG. 3, the perimeter of third rack assembly 126 may be defined by a wire framework 300. Wire framework 300 may include a floor 308 formed in combination with the floor portion of wire insert 302. Wire insert 302 may include at least two (2) of a hook 306, or the like, for removably coupling wire insert 302 to wire framework 300. The present example embodiment includes four (4) hooks 306. Wire insert 302 may be defined in the transverse direction T, between a first boundary wall 310 and a second boundary wall 312, whereupon hooks 306 may be positioned. In FIG. 3, wire insert 302 is shown coupled to wire framework 300 via hooks 306.

As shown in FIG. 4, generally, a caddy 400 may be inserted into third rack assembly 126. Caddy 400 may include a plurality of mounting projections 402, e.g., such as feet 402. Particularly, feet 402 may include tabs, hooks, snaps, and/or posts. Feet 402 may be positioned in a variety of positions as shown in FIGS. 4-6. Particularly, as shown in FIG. 4, feet 402 may be positioned along an outer edge 406 of a bottom side 404 of caddy 400. Generally, feet 402 may be spread out across bottom 404 of caddy 400, e.g., in both the lateral direction L and the transverse direction T. As may be seen in FIG. 5, feet 402 may include pair of feet spread across the transverse direction T for additional stability when placed in third rack assembly 126. As may be seen in

8

FIG. 6, in some additional or alternative embodiments, feet 402 may be positioned relative to a middle axis M, e.g., an axis that is centered in caddy 400 with respect to the lateral direction L of caddy 400. Particularly, in some example embodiments, feet 402 of caddy 400 may be disposed no more than three centimeters (3 cm) in the lateral direction L from the middle axis M of caddy 400. Feet 402 may be integrally molded with caddy 400 in example embodiments.

Shown in FIG. 7, a lower floor portion 316 of wire insert 302 may generally be configured to move beneath a third spray assembly 320. For example, wire insert 302 may be configured to translate within wire framework 300 and position lower floor portion 316 beneath, in the vertical direction V, third spray assembly 320. Wire insert 302 may include an upper floor portion 314 and the lower floor portion 316. The upper floor portion 314 and the lower floor portion 316 may be positioned at different heights with respect to the vertical direction V such that different size articles for washing may be placed upon wire insert 302. Lower floor portion 316 may be configured to interface, e.g., minimal gap, with floor 308 when wire insert 302 is inserted in wire framework 300. Thus, e.g., floor 308 may be nested against wire insert 302 when wire insert 302 is inserted in wire framework 300. Moreover, lower floor portion 316 may be positioned generally coplanar with floor 308 such that lower floor portion 316 and floor 308 may collectively form a portion of the floor of third rack assembly 126.

Moreover, caddy 400 may be configured to mount in a plurality of positions, e.g., position A, position B, or position C. As shown in FIG. 7, position A may be on top, or above, of third spray assembly 320. Position B may be on the rack wires of lower floor portion 316, and position C may be on the rack wires of upper floor portion 314. In general, feet 402 may be configured to engage with third spray assembly 320 and/or wires of third rack assembly 126. For example, feet 402 positioned at outer edge 406 of caddy 400 may engage with a perimeter P of third spray assembly 320. Additionally or alternatively, feet 402 positioned relative to middle axis M may be configured to interact with an inner edge I of third spray assembly 320. Further, feet 402 positioned at outer edge 406 of caddy 400 and feet 402 positioned relative to middle axis M may be configured to engage with the wires of upper floor portion 314 and lower floor portion 316 of third rack assembly 126.

As may be seen in FIGS. 8 and 9, third spray assembly 126 may include a plurality of posts 700, e.g., standoffs or protrusions, generally configured for restricting movement of caddy 400 in the lateral L or transverse T directions. Generally, posts 700 may be spread out across third spray assembly 320, e.g., in both the lateral direction L and the transverse direction T. In particular, as shown in FIG. 9, posts 700 may extend in the vertical direction V to pass through an opening 900 in caddy 400 and aid in restricting the movement of caddy 400.

As may be seen from the above, a dishwasher appliance may include a caddy generally configured to be positioned in multiple locations in the third rack assembly. The feet of the caddy may pass through the wires of the third rack assembly when the caddy is placed in the wire rack sections, and may sit on the perimeter of the third spray assembly when placed thereon. Additionally or alternatively, posts may be positioned on the third spray assembly that pass through the caddy in order to aid in positioning and restricting the movement of the caddy.

This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including

9

making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

What is claimed is:

1. A dishwasher appliance defining a vertical direction, a lateral direction, and a transverse direction, the vertical, lateral, and transverse directions being mutually perpendicular, the dishwasher appliance comprising:

a tub defining a wash chamber;

a wash conduit in the wash chamber;

a first rack assembly slidably positioned within the wash chamber;

a first spray assembly positioned in the wash chamber and configured to direct wash fluids at the first rack assembly;

a second rack assembly slidably positioned in the wash chamber above the first rack assembly;

a second spray assembly positioned in the wash chamber and configured to direct wash fluids at the second rack assembly;

a third rack assembly slidably positioned in the wash chamber above the second rack assembly, the third rack assembly comprising a wire framework defining a perimeter and a wire insert selectively mountable on the wire framework;

a third spray assembly positioned within the third rack assembly and configured to direct wash fluid at articles located in the third rack assembly; and

a caddy positionable on the third spray assembly, the caddy comprising a plurality of mounting projections at a bottom side of the caddy, the mounting projections configured to engage the third spray assembly in order to locate the caddy on the third spray assembly,

wherein the wire insert is slidably positionable within the wire framework beneath the third spray assembly, when the caddy is mounted to the third spray assembly.

2. The dishwasher appliance of claim 1, wherein the plurality of mounting projections of the caddy comprises one or more of tabs, hooks, snaps, and posts.

3. The dishwasher appliance of claim 1, wherein the third rack assembly comprises an upper rack portion and a lower rack portion, the upper rack portion disposed above the lower rack portion.

4. The dishwasher appliance of claim 3, wherein the plurality of mounting projections engages with wires of the third rack assembly when positioned on one of the upper rack portion and the lower rack portion.

5. The dishwasher appliance of claim 1, wherein the plurality of mounting projections of the caddy are disposed around an outer edge of the bottom side of the caddy.

6. The dishwasher appliance of claim 5, wherein the plurality of mounting projections at the outer edge of the caddy are configured to engage with a perimeter of the third spray assembly.

7. The dishwasher appliance of claim 1, wherein the plurality of mounting projections of the caddy are disposed no more than three centimeters in the lateral direction from a center axis of the caddy.

8. The dishwasher appliance of claim 7, wherein the plurality of mounting projections are configured to engage with an inner edge of the third spray assembly.

10

9. A dishwasher appliance defining a vertical direction, a lateral direction, and a transverse direction, the vertical, lateral, and transverse directions being mutually perpendicular, the dishwasher appliance comprising:

a tub defining a wash chamber;

a wash conduit in the wash chamber;

a first rack assembly slidably positioned within the wash chamber;

a first spray assembly positioned in the wash chamber and configured to direct wash fluids at the first rack assembly;

a second rack assembly slidably positioned in the wash chamber above the first rack assembly;

a second spray assembly positioned in the wash chamber and configured to direct wash fluids at the second rack assembly;

a third rack assembly slidably positioned in the wash chamber above the second rack assembly, the third rack assembly comprising a wire framework defining a perimeter and a wire insert selectively mountable on the wire framework;

a third spray assembly positioned within the third rack assembly and configured to direct wash fluid at articles located in the third rack assembly, the third spray assembly comprising a plurality of posts extending in the vertical direction; and

a caddy positionable on the third spray assembly, the caddy comprising a plurality of mounting projections at a bottom side of the caddy, the mounting projections configured to engage the third spray assembly in order to locate the caddy on the third spray assembly,

wherein the plurality of posts of the third spray assembly are configured to restrict movement of the caddy in both of the lateral direction and the transverse direction, and

wherein the wire insert is slidably positionable within the wire framework beneath the third spray assembly, when the caddy is mounted to the third spray assembly.

10. The dishwasher appliance of claim 9, wherein the plurality of mounting projections of the caddy comprises one or more of tabs, hooks, snaps, and posts.

11. The dishwasher appliance of claim 9, wherein the third rack assembly comprises an upper rack portion and a lower rack portion, the upper rack portion above the lower rack portion.

12. The dishwasher appliance of claim 11, wherein the plurality of mounting projections engages with wires of the third rack assembly when positioned on one of the upper rack portion and the lower rack portion.

13. The dishwasher appliance of claim 9, wherein the plurality of mounting projections of the caddy are disposed around an outer edge of the bottom side of the caddy.

14. The dishwasher appliance of claim 13, wherein the plurality of mounting projections at the outer edge of the caddy are configured to engage with a perimeter of the third spray assembly.

15. The dishwasher appliance of claim 9, wherein the plurality of mounting projections of the caddy are disposed no more than three centimeters in the lateral direction from a center axis of the caddy.

16. The dishwasher appliance of claim 15, wherein the plurality of mounting projections are configured to engage with an inner edge of the third spray assembly.

* * * * *